Academic year 2025-2026 (Odd Sem)

DEPARTMENT OF

Computer Science & Engineering

Date	7th Nov 2025	Maximum Marks	10+50
Course Code	CS354TA	Duration	120 Min
Sem-V	Test-1	Staff: HKK|ASP|SMS|CRM|MRA|PK	

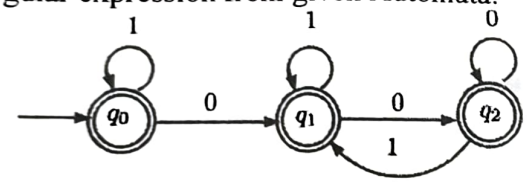
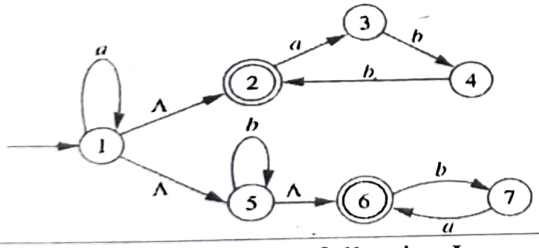
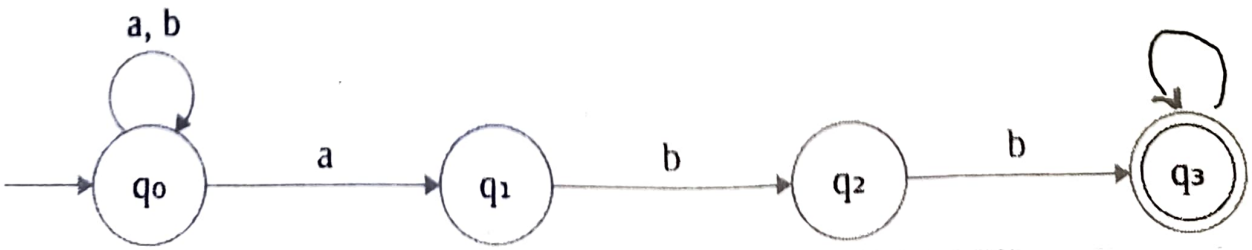
THEORY OF COMPUTATION

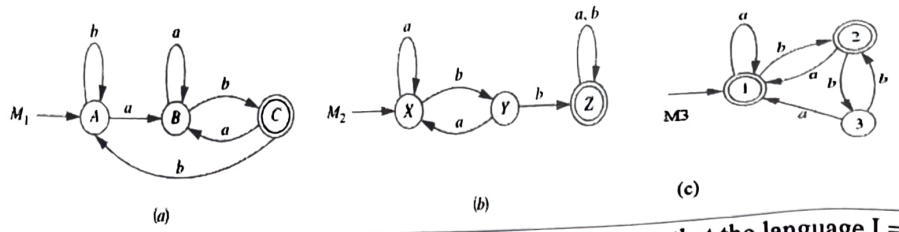
(Common to CSE & ISE)

PART-A

		Marks	BT	CO
1.1	Consider DFA A1 = ({q0,q1,q2} ,∑ ,δ,{q0,q1},q0) and DFA A2 = ({p01,p1,p2,p3} ,∑ ,δ,{p3},p0), then DFA of $(L(A1) \cap L(A2))^R$ has ____ final state(s).	1	3	2
1.2	How many strings of length less than 4 contains the language described by the regular expression (x+y)*y(a+ab)*?	1	2	1
1.3	Show that the language L= {vwv: v, w ∈ {a,b}*, |v|= 2} is regular	2	2	1
1.4	Find all strings in L((a + b) b (a + ab)*) of length less than four.	2	2	2
1.5	Construct Ɛ-NFA for the regular expressions a*(a+b)b*.	2	1	3
1.6	If regular expression R = (0 +1)*001(0 +1)* and homomorphism function h(0) = ab and h(1) = bc, Find h(R).	1	1	4
1.7	Define extended transition function for DFA.	1	1	3

PART-B

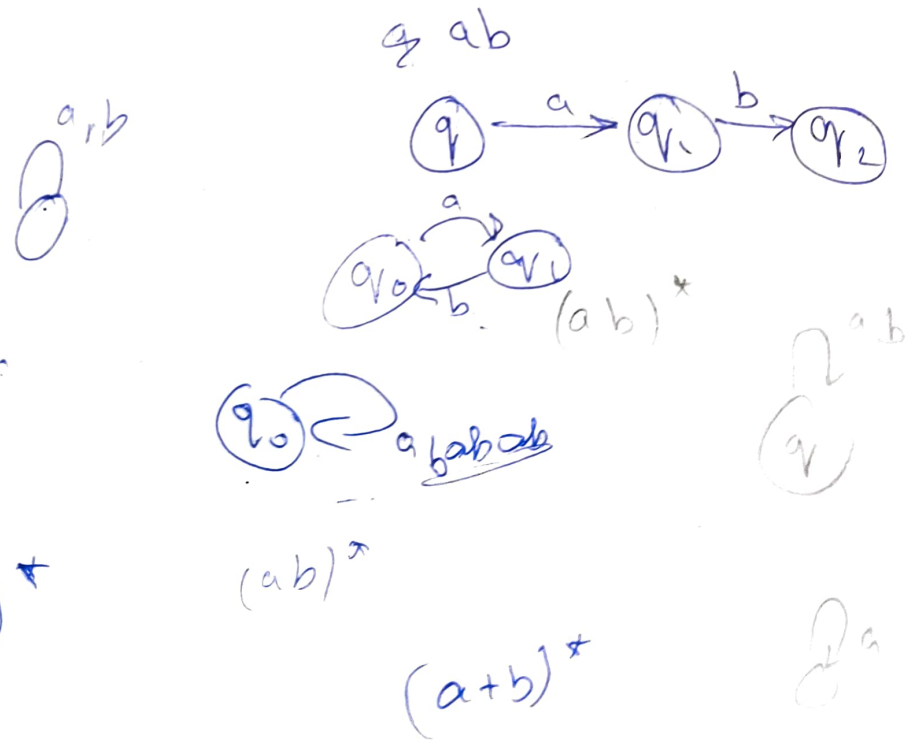
2.a	Find the regular expression from given Automata. 	6	3	3
2b	Describe decision algorithm to answer the following questions. i. Given a regular expression r and an DFA M, are the corresponding languages are same? ii. Given two NFA- ϵ do they accepts the same language?	4	2	1
3a	Consider the following Ɛ-NFA  a) Find Ɛ -Closure of each state b) Convert Ɛ-NFA to NFA	6	3	4
3b	Construct DFA for the following Languages over Σ = {a,b}, a. L1= w1aw2: |w1|≥ 3, |w2|≤ 5} b. L2= {ba^n: n ≥ 1,n≠ 5}.	4	2	
4a	Discuss the applications of regular expressions with an example for each	3	2	
4b	Construct DFA from the given NFA and minimize the DFA using Table filling algorithm. 	7	3	

5	Let M1, M2 and M3 are the DFA's pictured below recognizing languages L1, L2 and L3 respectively. Construct FAs accepting the following languages. i. L1 U L2 ii. L1 ∩ L3 iii. $L3^R$ iv. L2-L3 v. $\overline{L2}$'  	10	4	2
6.a	State and prove pumping lemma for regular Languages. Prove that the language L={ $a^n b^{2n}$, n>=1} is not regular	7	2	2
6.b	In each case below, find a string of minimum length in {a,b}* not in the language corresponding to the given regular expression. a. $b^*(ab)^*a^*$ b. $(a^* + b^*)(a^* + b^*)(a^* + b^*)$ c. $a^*(baa^*)^*b^*$ d. $b^*(a + ba)^*b^*$	3	1	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars	CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Max Marks	10	23	20	7	7	23	20	10	-	-

Course Outcomes:

1. Understand the fundamental concepts of theory of computations
2. Analyze the tools of finite automata to various fields of computer science.
3. Design solution model for complex problems, using the appropriate skills of automata theory for better results.
4. Apply automata skills in situations that describe computation effectively and efficiently.

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